

CS 598 Topics in LLM Agents: Summary of Agents for Retrieval-Augmented Generation (RAG)

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Problem and Motivation

Despite their strong text-generation capabilities, Large Language Models (LLMs) often falter in knowledge-intensive question-answering (QA) scenarios due to two critical shortcomings:

1. **Inefficient or Insufficient Retrieval for Varying Query Complexity:** Not all questions require the same retrieval depth. Some queries can be answered by leveraging the LLM’s internal (parametric) knowledge or with minimal external retrieval, while more complex queries demand multiple iterative lookups. Relying on a single, fixed retrieval strategy wastes computational resources for simpler queries and risks insufficient evidence collection for complex multi-hop questions. This mismatch motivates an *adaptive* retrieval approach—one that dynamically adjusts the retrieval method based on the perceived complexity of the query.
2. **Unreliable or Irrelevant Documents Leading to Hallucinations:** When the retrieval module returns inaccurate or irrelevant information, the LLM’s final answer is prone to hallucination and factual errors. Standard retrieval-augmented generation (RAG) frameworks rarely re-evaluate whether the retrieved evidence is correct before the LLM uses it to generate a response. As a result, even robust generation models produce misleading or incorrect outputs, highlighting the need for *corrective* mechanisms that can detect and fix flawed retrieval results in real time.

These two shortcomings motivate the development of methods like:

- **Adaptive-RAG**, which classifies each query by difficulty and selectively applies *no retrieval*, *single-step retrieval*, or *multi-step retrieval* to optimize both efficiency and effectiveness.
- **CRAG** (Corrective Retrieval Augmented Generation), which employs a lightweight evaluator to *assess* the quality of retrieved documents and correct retrieval failures (e.g., retrieving new evidence from the web if existing documents are deemed incorrect or ambiguous).

By dynamically tailoring retrieval strategies to query complexity and introducing corrective checks on retrieved content, these approaches significantly reduce computational overhead while improving the factual reliability of LLM-based QA.

Related Work

Adaptive Retrieval in LLMs

The concept of adaptive retrieval has received increasing attention, particularly for improving open-domain and multi-hop QA tasks. [8] propose a QA framework that employs a smaller LM-based classifier to predict query complexity and dynamically select retrieval strategies. In a related vein, [5] introduce Rowen, a framework leveraging consistency-based hallucination detection to determine when retrieval is necessary, thereby balancing parametric and non-parametric knowledge to reduce factual errors.

Beyond these individual frameworks, a recent survey by [6] offers a comprehensive overview of Retrieval-Augmented Generation (RAG) paradigms, highlighting adaptive retrieval techniques (e.g., FLARE and Self-RAG) and underscoring the complementary relationship between RAG and fine-tuning for knowledge-intensive tasks. Research on query difficulty classification—crucial for informing adaptive retrieval—includes [1], which reviews automatic methods for predicting question difficulty, and [17], which proposes an attention-based model to estimate difficulty by analyzing question components.

Earlier works in open-domain QA [3, 10] and multi-hop QA [18] laid the foundation for current adaptive retrieval approaches. [15] also explore strategies for adaptive retrieval, emphasizing trust and efficiency in LLM-driven QA systems.

Self-Reflection in LLMs

Recent studies have focused on leveraging self-reflection to enhance LLM response quality and factuality. Self-RAG [2] is a framework that enables a single LLM to perform on-demand retrieval and reflect on its own outputs using reflection tokens (ISREL, ISSUP, ISUSE). This approach significantly improves factual accuracy and citation reliability, outperforming earlier RAG-based methods on tasks like open-domain QA and fact verification. Another line of work explores critique-based refinement. [7] propose a method whereby the LLM validates and refines its outputs through tool interactions, addressing inconsistencies. Similarly, [16] investigate self-reflection in LLM problem-solving, demonstrating notable performance gains on more challenging queries—an outcome that aligns with the core aims of Self-RAG.

Additional techniques such as verification chains [4] and iterative refinement [14] have been shown to reduce hallucinations and enhance the reliability of generated responses. These strategies resonate with Self-RAG’s emphasis on using self-critique to improve both factual correctness and overall coherence.

Corrective Mechanisms in Retrieval-Augmented Generation

A distinct direction in RAG research focuses on corrective mechanisms to address retrieval failures, exemplified by Corrective Retrieval Augmented Generation (CRAG) proposed by [19]. Unlike Self-RAG’s internal reflection approach [2], CRAG introduces an external lightweight retrieval evaluator to assess document quality, triggering web searches to supplement or replace inadequate results. This method enhances robustness in knowledge-intensive tasks by dynamically correcting retrieval errors, a feature validated across diverse datasets. Complementary works include [11],

which integrates web searches to augment dialogue generation, though CRAG applies this specifically for error correction rather than general enhancement. Additionally, [9] explores active retrieval adjustments during generation, aligning with CRAG’s dynamic correction but lacking its explicit evaluator-driven approach. Another related effort by [13] employs iterative self-feedback to refine retrieval-augmented outputs, differing from CRAG’s focus on pre-generation correction via external sources. These studies collectively highlight a growing emphasis on mitigating retrieval shortcomings, with CRAG offering a plug-and-play solution that integrates seamlessly with existing RAG frameworks.

Efficiency and Concurrent RAG Approaches

Efficiency considerations in RAG-based systems have also been explored. [9] and [15] address runtime efficiency by selectively retrieving information and minimizing unnecessary external calls. Concurrently, [12] propose new training paradigms for retrieval augmentation, representing a departure from Self-RAG’s on-demand retrieval and self-reflection focus.

Taken together, these diverse efforts—ranging from adaptive retrieval strategies to self-reflection and efficiency optimizations—underscore the rapidly evolving landscape of Retrieval-Augmented Generation, informing the design choices and objectives behind both Adaptive-RAG and CRAG.

Solution Overview

Adaptive-RAG

To tackle the problem of being unable to select the most suitable strategy for retrieval-augmented LLMs based on query complexity, the authors propose **Adaptive-RAG**, a novel QA system that dynamically classifies query complexity and adapts retrieval strategies accordingly. Specifically, a smaller, sequence labeling-based classifier is trained to predict the complexity level of incoming queries. It does so using automatically collected labels derived from model predictions and dataset-provided inductive biases. Based on the classified complexity, the system selects an appropriate approach from three retrieval strategies:

- **No retrieval** for straightforward queries that LLMs can answer without external knowledge.
- **Single-step retrieval** for moderately complex queries requiring external knowledge from a single document.
- **Multi-step retrieval** for complex queries that require iterative retrieval and reasoning across multiple documents.

By adapting retrieval strategies based on query complexity, Adaptive-RAG minimizes unnecessary retrieval overhead for simple queries while enabling robust multi-hop reasoning for complex ones, striking a balance between efficiency and performance (Fig. 1).

Experimental results on multiple QA benchmarks demonstrate that Adaptive-RAG significantly improves both accuracy and efficiency over prior adaptive retrieval baselines, validating its effectiveness in real-world QA applications.

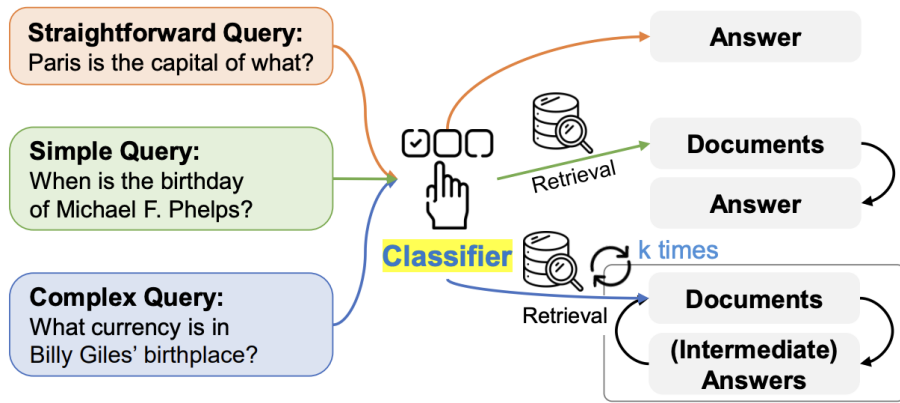


Figure 1: The system begins by classifying incoming questions based on their complexity—ranging from straightforward to simple or complex. Depending on this classification, it either performs a single retrieval step (for straightforward or simple queries) or multiple retrieval-and-answer iterations (for more complex queries). Each retrieval step surfaces relevant documents from which the model can generate an intermediate or final answer. By adapting the number of retrieval iterations according to query difficulty, Adaptive RAG efficiently handles a wide spectrum of questions while maintaining high answer quality.

CRAG

To address the issue of unreliable retrieved documents in RAG, the authors propose Corrective Retrieval Augmented Generation (CRAG). In CRAG, given an input query and the retrieved documents from any retriever, a lightweight retrieval evaluator is constructed to assess the relevance of the retrieved documents to the query. The evaluator quantifies the relevance into three confidence degrees, each triggering a corresponding action:

- **Correct:** If the retrieved documents are deemed relevant, they are refined through a knowledge decomposition and recomposition process to extract the most critical information while filtering out irrelevant content.
- **Incorrect:** If the retrieved documents are deemed irrelevant, they are discarded. Instead, large-scale web searches are conducted to retrieve more reliable external knowledge.
- **Ambiguous:** If the evaluator is uncertain, a hybrid approach is applied, combining refined internal knowledge from the retrieved documents with external web-searched knowledge.

This corrective mechanism ensures that the generation process is based on accurate and relevant information, thereby reducing hallucinations and improving the robustness of RAG-based approaches (Fig. 2). CRAG is designed as a plug-and-play solution that can be seamlessly integrated into existing RAG frameworks.

Experimental results on four datasets, *PopQA*, *Biography*, *PubHealth*, and *Arc-Challenge*, demonstrate that CRAG significantly enhances the performance of RAG-based approaches, making it more reliable for both short- and long-form generation tasks.

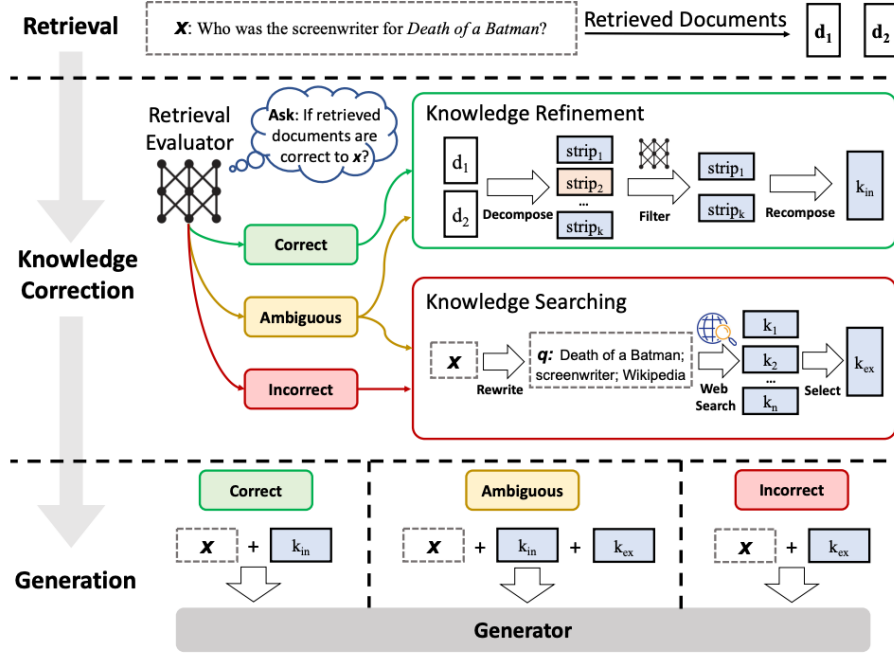


Figure 2: Given a query (e.g., “Who was the screenwriter for Death of a Batman?”), the system retrieves candidate documents and evaluates their correctness. If the documents are deemed correct, they go directly to the answer generator. Ambiguous documents undergo knowledge refinement (decomposition, filtering, and recombination), while incorrect documents prompt additional knowledge searching. The refined or corrected information is then passed to the generator to produce the final answer, ensuring robust handling of various retrieval outcomes.

Limitations

Adaptive-RAG

- The classifier is trained using automatically generated labels derived from model performance and dataset biases, which may introduce incorrect labels and affect accuracy.
- While Adaptive-RAG aims to optimize efficiency, it still incurs latency costs when handling complex multi-hop queries. For example, if the classifier misclassifies a single-step query to be multi-step, resources are wasted in doing extra retrieval.
- The classifier assigns each query to one of three complexity levels, but real-world queries may exist on a spectrum of complexity rather than distinct categories. It is not generalizable yet.

CRAG

1. CRAG requires a fine-tuned retrieval evaluator to assess the quality of retrieved documents, which introduces an additional training step and computational overhead.
2. The retrieval step may fail to retrieve and may misclassify relevant or irrelevant documents, leading to suboptimal correction cycles. The system relies on a threshold-based classification (*Correct*, *Incorrect*, *Ambiguous*), which may not always be precise.

3. Its effectiveness depends on the retrieval and generation models it is integrated with. Performance may vary significantly across different datasets and LLM architectures.
4. If retrieval results are deemed incorrect, CRAG uses web searches to fetch better information. However, web search results may introduce biased or unreliable information. Filtering and selecting high-quality sources (e.g., prioritizing Wikipedia) only partially mitigates this issue.

Future Research Directions

1. Develop a more comprehensive, balanced and accurate dataset for training the classifier in Adaptive-RAG.
2. Further refinement of the difficulty classifier. Implement multi-tier complexity modeling rather than a fixed three-class system to capture more nuanced differences. Reduce retrieval overhead by developing more adaptive stopping criteria to decide when retrieval should terminate.
3. For CRAG, investigate domain-specific adaptation for knowledge sources and reflection strategies to reduce hallucination and minimize repeated correction loops.
4. We can try eliminating the evaluator while maintaining accuracy, such as fine-tuning LLMs to predict the reliability of retrieved documents without external evaluation.
5. We can equip LLMs with better retrieval evaluation capabilities, such as implementing multi-level confidence scoring instead of fixed thresholds.
6. Another area of improvement includes better retrieval modeling to handle a wider range of queries, domain-specific complexities, and new knowledge sources.

Summary of Class Discussion

Q1. Is the evaluator a tool?

Not specifically for this paper, but it has the potential to be developed into one. It would be invoked when the model is uncertain about its answer. However, at its current stage, it is not generalizable enough to function as a standalone tool. Significant improvements would be required for broader applicability.

Q2. Are there any validation steps for the classification results of the LLMs? (This question is for both paper, as one classifies whether to do a retrieval and steps, the other for correct, incorrect, and ambiguous)

Currently, they take it as ground truths, but some future work could be done to improve this.

Q3. How large is the evaluator? Is it large enough?

These two models are considered large, but they are tuned on a small dataset. The authors did not show the performance without fine-tuning; we suspect that without fine-tuning, it does not perform well.

Q4. In the confusion matrix in Figure 3, 23% of one is misclassified as multi, and 47% of no as one. It feels like the classifier is more likely to predict the query with higher complexity than the ground truth. Also, the proposed method did not outperform the single-step baseline on single-step tasks. What are your thoughts on this?

This is like a trade-off between precision and recall. Predicting one or more instead of no will result in better performance as the model will be more confident in the answer, but this also causes the method to become less efficient.

Q5. In the CRAG paper, they first filtered the documents based on a trained classifier, and then they broke the document down into strips and again filtered it. What if we remove the first step and just break down the document and filter the strips directly? This would give a better fine-grained control of the quality.

True, your point makes sense. That is one way to go about it as well. It should probably work well.

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